NURTUREPATH AI / 01 / MVP

Codex Master Prompt: Build the Complete MVP

Full implementation specification | Responsive web application | Offline-first maternal care workflow

Prepared as a practical implementation and review document. AI-generated clinical content must not be used as medical advice without qualified review.

00. MASTER CODEX INSTRUCTION

You are Codex acting as a senior full-stack engineer, product designer, QA engineer and security-minded technical lead. Build a complete, runnable, responsive NurturePath AI MVP in the current repository. Do not stop at scaffolding, mock-only screens or a plan. Inspect existing code first; preserve functioning work; implement incrementally; run tests and document results. If a secret or external provider is unavailable, implement a fully functional local simulator and a clearly labeled integration adapter. Never fabricate real clinical validation or patient data.

Definition of done: a developer can clone, configure .env.example, install dependencies, migrate/seed, run locally, sign in with demo roles, complete an end-to-end maternal follow-up and referral workflow, see a measurable dashboard, run automated tests and deploy a demo safely.

01. PRODUCT, AUDIENCE AND NON-NEGOTIABLE BOUNDARIES

Product: NurturePath AI. Tagline: Every mother deserves care that follows her home. Ghana-first, international-ready, offline-first maternal care follow-up and referral coordination. Users: community health workers (CHWs), midwives/clinic coordinators, programme administrators and consenting adult mothers. The system is NOT an electronic medical record, diagnostic engine, emergency dispatch service or substitute for licensed care.

MVP scope: adult maternal users only; prenatal and postnatal visit scheduling, follow-up tasks, consent, approved educational messages, referral creation and closure, offline capture, language-ready UI, role-aware aggregate analytics. Do not build medical risk prediction, symptom diagnosis, medication recommendations, autonomous triage, public patient search or child accounts.

For warning signs, use clinician-approved fixed safety copy and direct users to local emergency services/clinics; never let the LLM decide emergency severity. All pilot data should be synthetic unless a real health partner and documented lawful data-processing arrangements exist.

02. TECHNICAL STACK AND REPOSITORY

Use TypeScript throughout; Next.js App Router + React + Tailwind CSS + accessible component primitives (shadcn/ui or equivalent), Zod validation, React Hook Form, PostgreSQL with Prisma, Auth.js or an equally well-maintained session solution, TanStack Query as needed, IndexedDB via Dexie for offline queue, Vitest and Testing Library for unit/component tests, Playwright for end-to-end, ESLint/Prettier, Docker Compose for local PostgreSQL. Prefer boring, maintained libraries and pin versions in lockfile.

Repository layout: app/(public), app/(auth), app/(dashboard)/worker, /clinic, /admin; components/ui, components/features; lib/auth, db, permissions, consent, encryption, audit, ai, messaging, offline, analytics; prisma/schema.prisma, seed.ts; tests/unit, integration, e2e; docs/architecture.md, privacy.md, safety.md, deployment.md, api.md; .env.example; README.md. Add GitHub Actions CI for lint, typecheck, tests and build.

03. BRAND, LOGO AND DESIGN SYSTEM

Brand identity: NurturePath AI; approachable, trustworthy, dignified, non-stereotyped. Primary deep navy #123047; primary teal #087E83; pale mint #E9F7F4; warm coral accent #F4A58A; white #FFFFFF; text slate #334A57; danger #B42318. Use coral sparingly, never as sole indicator. Logo: original minimalist continuous-line heart enclosing a small upward care path/ECG stroke, with a rounded sans wordmark 'NurturePath' and small 'AI'; create SVG assets in /public/brand/logo.svg, mark.svg, favicon.svg and social preview. Do not copy UNICEF or medical organization branding.

Typography: Inter or locally bundled equivalent; 16px base, 1.5 line height; large tap targets >=44px; 4.5:1 normal text contrast; WCAG 2.2 AA goal. Responsive breakpoints 360, 390, 768, 1024, 1440; mobile bottom navigation for worker tasks, desktop sidebar; never require horizontal scrolling for forms or tables; support keyboard, screen readers, reduced motion, visible focus, error summaries and accessible charts. Provide light theme first, optional dark theme.

04. ROLES AND PERMISSIONS

Roles: PLATFORM_ADMIN (tenant setup, aggregate reporting, no routine access to sensitive records), CLINIC_ADMIN (staff assignment, care coordination within assigned clinic), HEALTH_WORKER (assigned patient records, visit tasks, referrals), MOTHER_PORTAL (optional limited own appointments and consent), AUDITOR (metadata/audit read only, no clinical content). Enforce permissions on server/API and database query scope, not only in UI. Tenant_id and clinic_id on all appropriate records. No cross-tenant data leakage. Restrict exports and require explicit authorization.

05. CORE USER JOURNEYS

Journey A: worker logs in, chooses clinic, registers a consenting adult mother using minimum data, selects language/contact preferences, records estimated due date or postpartum status, schedules next visit and receives confirmation. Journey B: worker opens Today dashboard, sees due/overdue follow-ups, opens record, logs contact outcome (reached, no answer, rescheduled, declined), creates next action. Journey C: worker creates referral with reason selected from controlled non-diagnostic categories, receiving facility, urgency field only when clinician-selected, records referral status (created, accepted, attended, closed, unable to contact) and closure notes. Journey D: clinic admin reviews anonymized metrics and task completion; admin cannot silently read all patient details.

Journey E: mother can opt in/out of contact channels, review appointments and request a call-back; never require smartphone ownership. Demonstrate SMS/WhatsApp adapters in sandbox mode with message previews and simulated delivery logs; only enable live messaging when configured and lawful.

06. REQUIRED PAGES AND COMPONENTS

Public: home with concise problem/solution, how-it-works, safety/privacy explanation, partner enquiry, accessibility and legal pages. Auth: login, password reset, role-specific redirect. Worker: overview with overdue and due tasks, searchable assigned mother list, registration wizard, mother detail timeline, appointment scheduler, follow-up log, referral form and referral status board, offline sync indicator. Clinic: staff assignment, facility management, task queue, aggregate impact dashboard. Admin: tenant/clinic management, access logs, configuration and approved message templates. Mother portal: minimal appointment and consent view if feasible after worker flow is complete.

Provide skeletons, loading/empty/error states, optimistic action rollback, accessible toast, confirmation dialogs for destructive actions, pagination and filters. Design all pages with mobile-first layouts, polished realistic synthetic demo data and consistent spacing.

07. DATABASE MODEL AND API CONTRACT

Prisma entities: Organization(id,name,country,createdAt); Clinic(id,organizationId,name,region); User(id,organizationId,clinicId?,role,name,email,passwordHash?,status); Mother(id,organizationId,clinicId,assignedWorkerId,displayCode,name?,phoneEncrypted?,preferredLanguage,careStage,estimatedDueDate?,birthDate?,createdAt,archivedAt?);
Consent(id,motherId,purpose,channel,version,grantedAt,withdrawnAt,recordedBy);
Appointment(id,motherId,clinicId,scheduledAt,status,createdBy);
FollowUp(id,motherId,appointmentId?,dueAt,status,outcome,assignedTo,completedAt);
Referral(id,motherId,clinicId,receivingFacility,reasonCode,status,createdAt,closedAt);
MessageTemplate(id,locale,category,approvedBy,approvedAt,body,version);
MessageDelivery(id,motherId,templateId,channel,status,providerId?,createdAt);
AuditEvent(id,actorId,organizationId,entityType,entityId,action,timestamp,metadataRedacted);
OfflineMutation(idempotencyKey,deviceId,actorId,mutationType,payloadHash,processedAt). Include indexes, constraints, enums, migrations and soft-delete/retention rules.

API: GET/POST /api/mothers; GET/PATCH /api/mothers/:id; POST /api/consents; GET/POST /api/appointments; GET/POST/PATCH /api/followups; GET/POST/PATCH /api/referrals; GET /api/dashboard/metrics; POST /api/offline/sync; GET /api/audit (authorized); POST /api/messages/preview; POST /api/ai/simplify-approved-content. Use typed request/response DTOs, Zod validation, rate limits, pagination, audit logs and consistent error envelopes. Never return encrypted phone fields to unauthorized users.

08. AI AND LANGUAGE SAFETY

AI MVP task is limited to transforming pre-approved, non-personalized educational copy into simpler language or draft translations. Use a provider adapter and a deterministic offline fallback; retrieve only versioned clinician-approved source documents. Never include identifiable patient details in prompts. Reject prompts asking for diagnosis, treatment, medication, pregnancy risk scoring or emergency triage. Show 'AI-assisted; reviewed content only' label and human approval before publishing generated templates. Log model/version, source IDs, approval and errors without logging PHI. Support English at launch, design locale keys for Twi, Ewe, Dagbani and other languages, but do not claim those translations are validated until human-reviewed.

09. OFFLINE-FIRST AND SYNCHRONIZATION

Installable PWA: cache application shell and safe static assets; IndexedDB queue for encrypted-at-rest-on-device or minimally sensitive pending mutations; no sensitive API response in service-worker shared cache; offline banner, pending-count and explicit Sync button. Queue writes with UUID idempotency keys, timestamps and version numbers. Server validates auth/tenant on replay and handles duplicate requests. Define conflicts: server-authoritative for permissions/consent withdrawal; show explicit review for conflicting appointment edits; never overwrite withdrawn consent with stale offline data. Document browser-storage limitations and wipe local records on logout. Demonstrate airplane-mode registration with synthetic data and reconnection sync.

10. PRIVACY, SECURITY AND HEALTH SAFETY

Explicit, granular consent for registration, reminders and messaging channels; opt-out and withdrawal; data minimization; avoid collecting diagnosis, national ID or exact location for MVP. Use TLS in deployment, secure httpOnly SameSite cookies, password hashing, CSRF protections as applicable, rate limiting, secret management, input validation, least privilege, security headers, audit logs, backup encryption and retention schedule. Protect phone numbers at application level using server-managed encryption keys; separate pseudonymous analytics. Implement Ghana Data Protection Act compliance assessment and consult a qualified professional before processing real patient data. No real-patient launch without a lawful basis, agreements, DPIA/risk assessment where appropriate, incident response, clinical review and partner authorization.

Safety UX: clear not-for-emergencies banner; 'Seek urgent medical care' link/text for danger signs using static clinician-reviewed copy; no automatic escalation promises. Make referral completion status explicit rather than suggesting a referral was fulfilled when it was merely created.

11. ANALYTICS AND IMPACT

Dashboard: active enrolled mothers, appointments scheduled, due and overdue, follow-up contact rate, referrals created, referral closure rate, opt-out rate and median time to contact. Include denominators, date ranges, missing-data indicators and small-number suppression (e.g., suppress cells under 10 in public outputs). Separate synthetic demo metrics from pilot results with visible DEMO label. Exports must be aggregated and privacy-reviewed; never publish personal or sensitive health data.

12. SEED DATA, TESTS AND ACCEPTANCE CRITERIA

Seed one organization, two clinics, role-based demo accounts and 20 clearly fictional mothers; create a demo reset script restricted to development. Automated tests: role isolation, consent withdrawal stops messages, encrypted contact handling, offline duplicate replay, appointment overdue calculation, referral status transitions, no PHI in AI prompts, no cross-tenant API access, keyboard form submission, mobile viewport overflow. E2E golden path: login as worker -> register synthetic mother -> record consent -> schedule visit -> mark overdue -> follow up -> refer -> close referral -> clinic dashboard updates. Add negative cases and CI.

Deliver complete source, migrations, working seed, .env.example, docker-compose.yml, README quickstart, screenshots, API docs, safety and threat model docs, and a demo script. Provide exact commands and real test outputs in final summary. Do not claim tests pass unless run. No placeholder buttons or dead routes in critical flows.

13. BUILD ORDER AND CODEX EXECUTION RULES

Phase 1: repo audit, design tokens, logo SVG, auth, database, migrations, seed. Phase 2: worker dashboard, mother registration, consent, appointments, follow-ups. Phase 3: referrals, clinic dashboards, audit, privacy. Phase 4: offline sync, messaging simulator, approved-content AI adapter. Phase 5: responsive/accessibility testing, CI, documentation and demo deployment. At each phase: implement, run lint/typecheck/tests, fix regressions, commit logically. Ask only when blocked by legal/clinical decisions or unavailable credentials; otherwise choose safe defaults and record assumptions.